

Should I Get Gas?

An AI-powered gas price prediction and timing advisor — not just where prices are, but **when** to fill up.

The Idea

Most gas price apps tell you *where* to find cheap gas right now. This project does something different: it tells you **whether today is a good day to fill up** — and if not, when you should.

The site analyzes historical price trends, crude oil futures, refinery news, geopolitical events, and seasonal patterns to generate a plain-English recommendation like:

- *"Prices are near a weekly low. Good day to fill up."*
- *"Prices are rising and likely to peak mid-week. Consider getting a little today and waiting until Thursday for a full tank."*
- *"Regional supply disruption detected. Prices may spike — fill up now if you can."*

You open the site, get an instant recommendation, and move on.

Core Features (v1 Scope)

Instant, Location-Aware Recommendations

- On load, the site detects your **approximate location from your IP address** (no permission prompt) to get state/regional gas prices
- Displays the current average price for your region and a simple **Buy / Wait / Partial** recommendation
- A single "Use My Exact Location" button unlocks nearby station prices (browser geolocation, opt-in)
- If a user wants granular station-level data, redirect to GasBuddy — we send them there rather than duplicate their core feature

AI Timing Advisor

- Predicts whether prices in your region are likely to go **up, down, or stay flat** over the next 2–5 days

- Renders a small price-trend sparkline showing historical context (last 2 weeks vs. today)
- Explains *why* the recommendation was made: "Crude oil dropped 3% this week" or "Summer blend switchover driving refinery costs up"

Context Panel

- Current crude oil price (WTI and Brent) and direction
- Relevant news headlines affecting gas prices (geopolitical events, OPEC decisions, refinery outages, hurricane season)
- Breakdown of what's driving today's price: crude oil, refining, taxes, distribution

Research Findings

Domain

`shouldigetgas.com` — DNS returns NXDOMAIN, indicating the domain is likely available for registration.

Competitive Landscape

Existing gas apps focus on **real-time price comparison**, not prediction:

App	What it does	Gap
GasBuddy	Crowdsourced current prices by station	No timing advice
Waze / Google Maps	Shows nearby prices during navigation	No timing advice
Upside	Cash back after purchase	No price intel
AAA Mobile	Station finder	No timing advice

No major consumer-facing product answers "should I fill up today or wait?" This is the gap.

Data Sources

Primary — EIA Open Data API (Free, Official)

The U.S. Energy Information Administration publishes weekly retail gasoline prices by region and state, freely available via their [Open Data API](#). No cost, just an API key.

- Weekly average prices by state and metro area

- Historical data going back decades (ideal for training a prediction model)
- Also provides crude oil spot prices (WTI), refinery utilization, and import data
- Endpoint: <https://api.eia.gov/v2/petroleum/pri/gnd/data/>

Secondary — GasBuddy (No Official Public API)

GasBuddy does not offer a documented public API. Options include: - `py-gasbuddy` — an unofficial Python wrapper around GasBuddy's GraphQL API (updated May 2026) - Third-party scraping services (Apify, ScrapingBee) — check terms of service before use - Commercial data licensing — contact GasBuddy directly - **Recommended v1 approach:** Use EIA for price trends/predictions, deep-link users to GasBuddy for exact station prices

News & Sentiment

- [NewsAPI](#) or GDELT for real-time headlines on crude oil, OPEC, refinery disruptions
- Crude oil futures: EIA API, or [OilPriceAPI](#) (has a free tier)

IP Geolocation

- [ipapi.co](#) or MaxMind GeoIP2 — map IP address to state/country silently on page load
- No permission prompt; resolution is state-level (sufficient to pull EIA regional data)

How Gas Prices Work

Gas prices at the pump break down roughly as:

Component	Share
Crude oil	~61%
Refining costs	~14%
Federal + state taxes	~14%
Distribution & marketing	~11%

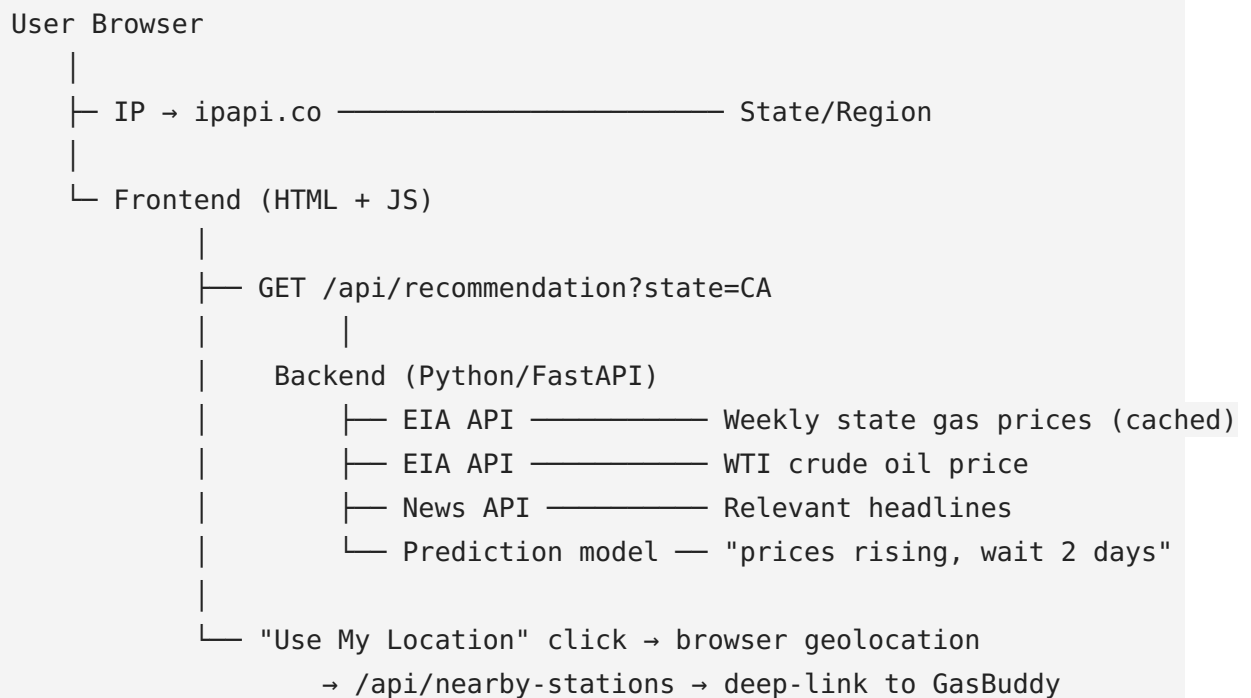
Key dynamics to model: - Crude oil changes pass through to pump prices within ~2 weeks (EIA data) - Prices vary significantly by state due to taxes and reformulation requirements - Seasonal patterns: prices typically rise in spring (summer blend switchover, higher demand) and fall in autumn - Geopolitical events (OPEC cuts, sanctions, conflicts) can cause rapid crude swings - Refinery outages (fire, maintenance, hurricanes) create regional spikes independent of crude prices

Prediction Approach

Academic research (2024–2025) shows LSTM (Long Short-Term Memory) neural networks achieve ~8.5% MAPE on 1-step gas price forecasting, outperforming simpler models. Hybrid CNN-LSTM-Attention architectures push accuracy further.

For v1, a simpler time-series approach (exponential smoothing, linear regression on recent EIA weekly data + crude oil trend) is likely sufficient to produce a directional "up / down / flat over next week" signal with reasonable confidence. Reserve ML complexity for v2.

Proposed Architecture



Potential Monetization (Later)

- Affiliate link to GasBuddy Pay / GasBuddy+ when redirecting users
 - GasBuddy commercial data license (if volume warrants)
 - Optional: a "Gas Budget Tracker" feature (enter your car's tank size, get weekly spend estimates)
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Tech Stack (Proposed)

Layer	Choice	Why
Frontend	Vanilla HTML/CSS/JS	Zero build complexity for v1
Backend	Python + FastAPI	Fast to iterate, great data science libs
Prediction	<code>statsmodels</code> / <code>scikit-learn</code>	Sufficient for v1 trend detection
Data cache	Redis or SQLite	Cache EIA weekly data, avoid rate limits
Hosting	Railway / Render / Fly.io	Free tier, simple deploy
IP Geo	ipapi.co	Free tier, no signup for basic use

What This Is Not

- Not a replacement for GasBuddy's station-level data (we'll link out)
- Not a commodity trading signal (we're predicting retail pump prices, not crude futures)
- Not a navigation app

Status

Early prototype / research phase. No code yet — this README is the starting point.

Next steps: 1. Register domain (if available — DNS check suggests it is) 2. Prototype EIA API integration and regional price fetch 3. Build minimal frontend with IP geolocation → state price display 4. Add simple week-over-week trend indicator 5. Iterate on the prediction model

Sources & References

- [EIA Open Data API](#) — free weekly gas price data by state/region
- [EIA Factors Affecting Gasoline Prices](#)
- [EIA Gasoline & Diesel Weekly Update](#)
- [American Petroleum Institute — How Gas Prices Are Determined](#)
- [py-gasbuddy \(PyPI\)](#) — unofficial GasBuddy GraphQL wrapper

- [PMC — Natural Gas Price Prediction Using AI Models](#)
- [GitHub — Gas Price ML Analysis](#)